

Module 5 Lab 3: Textbook Problem 5.5

CS 82A - Introduction to Data Science

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GitHub: https://github.com/ramishatasfia5156-commits/cs82a-coursework/blob/main/module5/lab3/module5_lab3.ipynb

Load the Data

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="whitegrid")

cust = pd.read_csv("custdata.tsv", sep="\t")
cols = ["age", "income", "num.vehicles"]
print("Shape as loaded:", cust.shape)
print()
print(cust[cols].describe().round(1))
```

Shape as loaded: (1000, 11)

	age	income	num.vehicles
count	1000.0	1000.0	944.0
mean	51.7	53504.8	1.9
std	18.9	65478.1	1.1
min	0.0	-8700.0	0.0
25%	38.0	14600.0	1.0
50%	50.0	35000.0	2.0
75%	64.0	67000.0	2.0
max	146.7	615000.0	6.0

BEFORE Cleaning: Correlation Matrix (Wrong on Purpose)

```
In [2]: # pandas silently skips the 56 missing vehicle counts pair by pair,
# but the impossible ages and negative income are all still included
corr_before = cust[cols].corr()
print("BEFORE cleaning (invalid values included):")
print(corr_before.round(3))
```

BEFORE cleaning (invalid values included):

	age	income	num.vehicles
age	1.000	0.027	-0.064
income	0.027	1.000	0.139
num.vehicles	-0.064	0.139	1.000

Find the Invalid Values

```
In [3]: bad_age = (cust["age"] <= 0) | (cust["age"] > 110) # impossible ages
bad_income = cust["income"] < 0 # negative income
missing_vehicles = cust["num.vehicles"].isna() # no vehicle count

print("Age = 0:", (cust["age"] == 0).sum())
print("Age > 110:", (cust["age"] > 110).sum())
print(sorted(cust.loc[cust["age"] > 110, "age"].round(1)))
print("Income < 0:", bad_income.sum(),
      list(cust.loc[bad_income, "income"]))
print("Missing num.vehicles:", missing_vehicles.sum())

invalid = bad_age | bad_income | missing_vehicles
print()
print("Total rows removed:", invalid.sum())
```

Age = 0: 3

Age > 110: 8

[113.3, 123.1, 124.8, 126.7, 130.4, 136.1, 137.7, 146.7]

Income < 0: 1 [-8700]

Missing num.vehicles: 56

Total rows removed: 68

```
In [4]: clean = cust[~invalid]
print("Rows before:", len(cust))
print("Rows after:", len(clean))
print("Removed:", len(cust) - len(clean))
print()
print(clean[cols].describe().round(1))
```

Rows before: 1000

Rows after: 932

Removed: 68

	age	income	num.vehicles
count	932.0	932.0	932.0
mean	51.4	55949.3	1.9
std	16.8	66040.1	1.1
min	19.0	0.0	0.0
25%	39.0	17000.0	1.0
50%	50.0	37000.0	2.0
75%	63.0	70000.0	2.0
max	93.0	615000.0	6.0

What I Removed and Why

I eliminated 68 cases prior to determining the clean correlation coefficients due to one of two factors: either there was likely an error in the data entry/management process; or the value(s) were blank. There were three customers who listed their ages as zero and 8 others whose ages ranged from 113 through 147 (including the decimals). Zero and ages above 110 are each impossible ages for actual customers. Only one customer had an income less than zero (-\$8,700) which is also not a valid income. The remaining 56 customers did not have a vehicle count recorded and therefore were unable to participate in a vehicle correlation. No cases were deleted twice by this process leaving me with the 932 case records of the original 1000.

AFTER Cleaning: Correlation Matrix

```
In [5]: corr_after = clean[cols].corr()
print("AFTER cleaning (932 valid rows):")
print(corr_after.round(3))
```

```
AFTER cleaning (932 valid rows):
          age  income  num.vehicles
age         1.000  -0.003        -0.066
income      -0.003   1.000         0.138
num.vehicles -0.066   0.138         1.000
```

Before vs. After

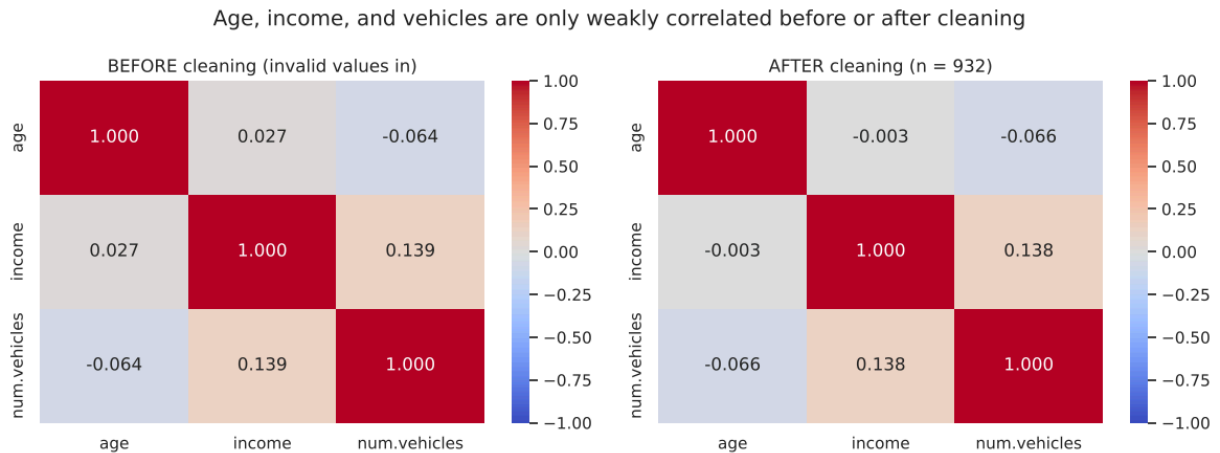
```
In [6]: pairs = [("age", "income"), ("age", "num.vehicles"),
                ("income", "num.vehicles")]
comparison = pd.DataFrame({
    "pair": [f"{a} vs {b}" for a, b in pairs],
    "r_before": [corr_before.loc[a, b] for a, b in pairs],
    "r_after": [corr_after.loc[a, b] for a, b in pairs],
}).round(3)
comparison
```

```
Out[6]:
```

	pair	r_before	r_after
0	age vs income	0.027	-0.003
1	age vs num.vehicles	-0.064	-0.066
2	income vs num.vehicles	0.139	0.138

```
In [7]: fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))
for ax, corr, title in [
    (axes[0], corr_before, "BEFORE cleaning (invalid values in)"),
    (axes[1], corr_after, "AFTER cleaning (n = 932)"]]:
    sns.heatmap(corr, annot=True, fmt=".3f", cmap="coolwarm",
                vmin=-1, vmax=1, ax=ax)
    ax.set_title(title)
fig.suptitle("Age, income, and vehicles are only weakly correlated "
            "before or after cleaning")
```

```
plt.tight_layout()
plt.show()
```



Interpretation (Cleaned Data Only)

After being cleaned there is virtually no relationship among all three variables. The only variable pair that showed a slight relationship were the number of vehicles owned by a customer versus their income which was a positive, but weak relationship ($r = .14$) indicating that those who earned a higher income tended to own slightly more vehicles than other customers. A very weak negative relationship existed between the age of the customer and the number of vehicles they owned ($r = -.07$). An almost non-existent relationship existed between the age of the customer and their income ($r = -.003$).

AI Disclosure

Claude (Anthropic) was used to help explain concepts and draft the code for this lab. All code was reviewed and understood by me, and I wrote the written responses in my own words. I can explain why ages of 0 or above 110 and negative incomes are invalid, why the rows with no vehicle count were removed, how the before and after correlation matrices differ, and why only the cleaned result should be interpreted.

Key prompt: "Help me with Module 5 Lab 3: Textbook Problem 5.5 using the attached data."